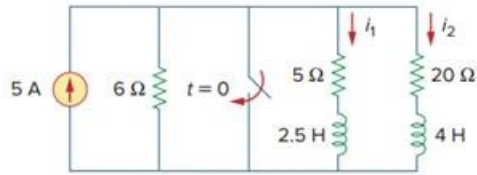


Problem

Find $i_1(t)$ and $i_2(t)$ for $t > 0$ in the circuit of Fig. 7.123.

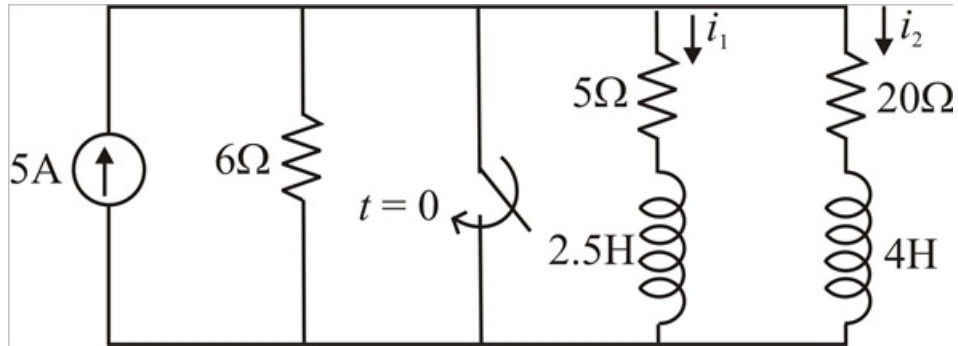
Fig. 7.123:



Step-by-step solution

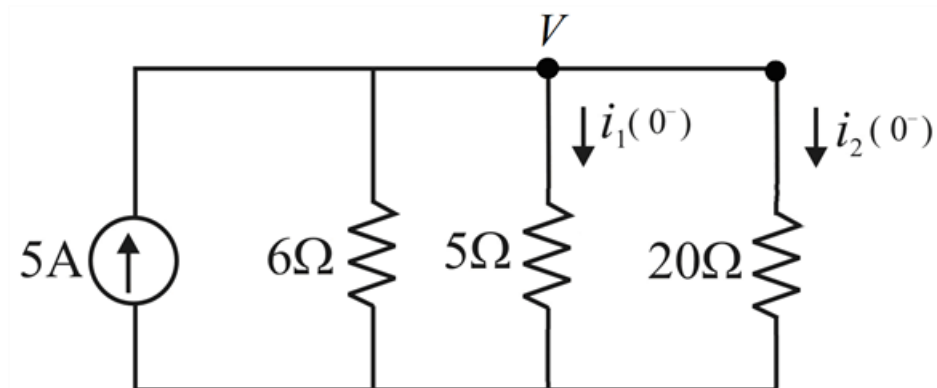
Step 1 of 14

Given circuit diagram



Step 2 of 14

Switch is closed at $t = 0$, for $t < 0$ inductors acts like short circuit at $t = 0^-$ and the circuit becomes as below



Step 3 of 14

Assume node voltage as V as shown above

In the above circuit 6Ω , 5Ω and 20Ω are in parallel

$$R_{eq} = 6 \parallel 5 \parallel 20$$

$$\frac{1}{R_{eq}} = \frac{1}{6} + \frac{1}{5} + \frac{1}{20}$$

Step 4 of 14

$$R_{eq} = 6 \parallel 5 \parallel 20$$

$$\frac{1}{R_{eq}} = \frac{1}{6} + \frac{1}{5} + \frac{1}{20}$$

$$R_{eq} = 2.4$$

$$V = 2.4 \times 5$$

$$V = 12 \text{ V}$$

Step 5 of 14

$$\therefore i_1(0^-) = \frac{12}{5}$$

$$i_1(0^-) = 2.4 \text{ A}$$

$$i_2(0^-) = \frac{12}{20}$$

$$i_2(0^-) = 0.6 \text{ A}$$

Step 6 of 14

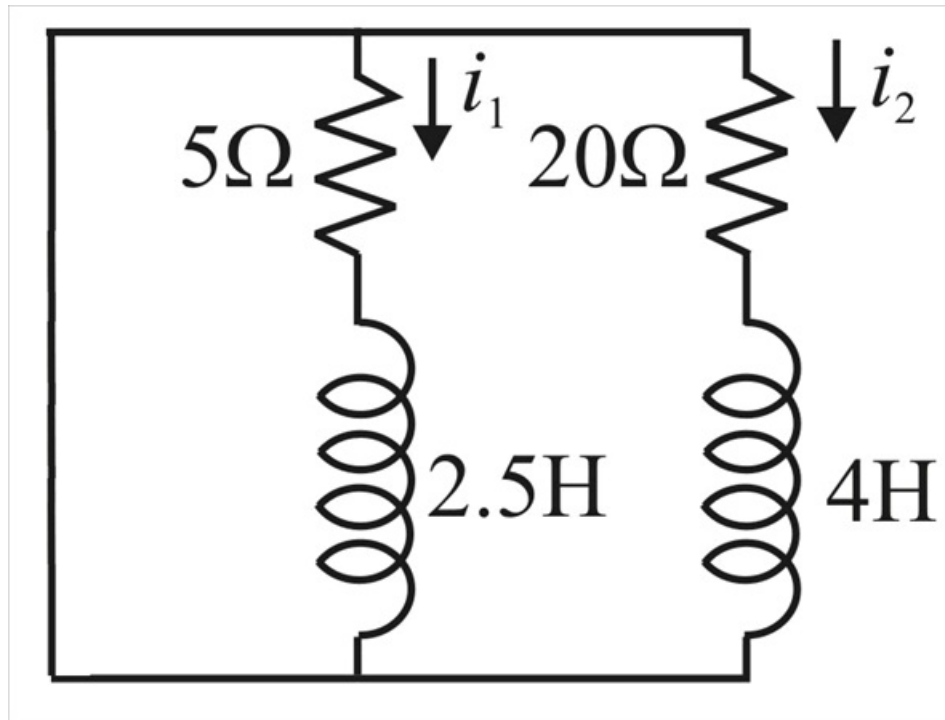
In inductor current can't change instantaneously

$$i_1(0^-) = i_1(0^+) = i_1(0) = 2.4 \text{ A}$$

$$i_2(0^-) = i_2(0^+) = i_2(0) = 0.6 \text{ A}$$

Step 7 of 14

Switch is closed at $t = 0$, so for $t > 0$ circuit is



Step 8 of 14

Above circuit is source free network

$$i_1(\infty) = 0$$

$$i_2(\infty) = 0$$

$\therefore$ First inductor current

$$i_1(t) = i_1(\infty) + [i_1(0) - i_1(\infty)]e^{-\frac{t}{\tau_1}}$$

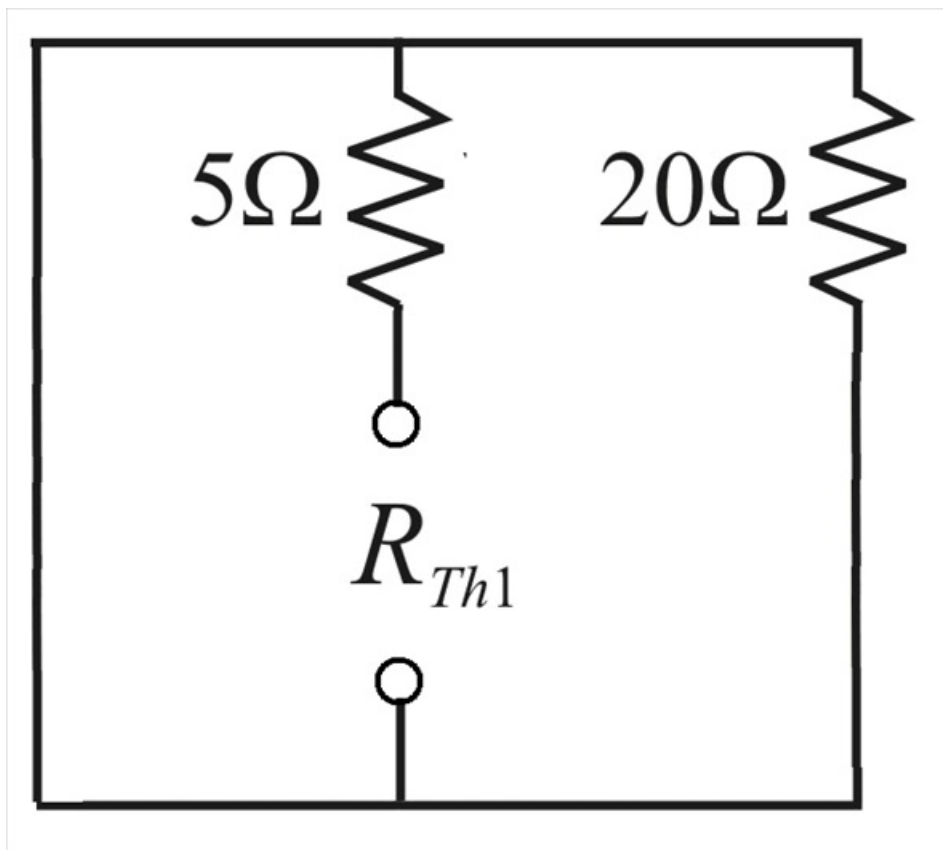
$$i_1(t) = 0 + (2.4 - 0)e^{-\frac{t}{\tau_1}}$$

$$i_1(t) = 2.4e^{-\frac{t}{\tau_1}} \text{ for } t > 0$$

Here $\tau_1 = \frac{L_1}{R_{Th1}}$

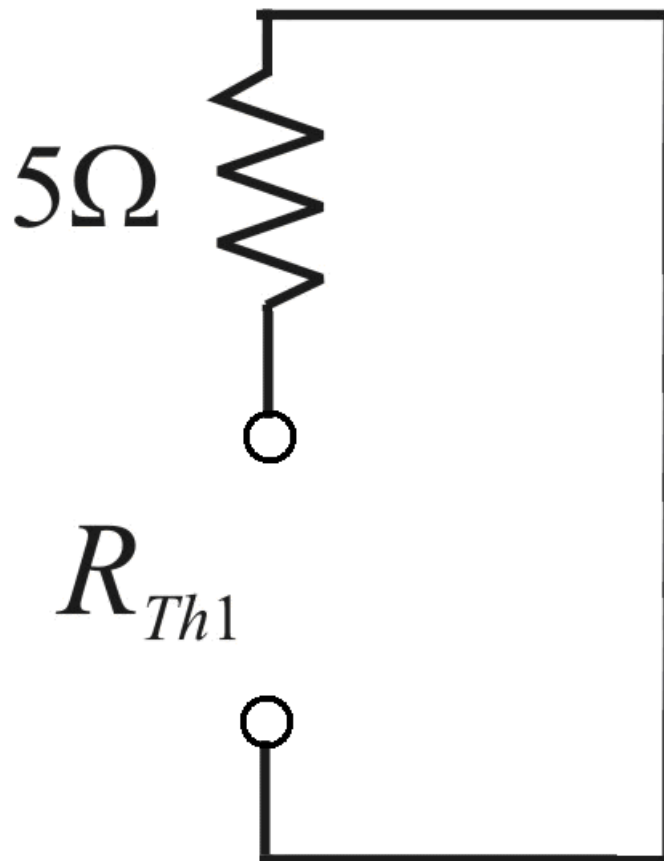
R_{Th1} is Thevenin resistance across inductor 2.5H by replacing inductor 4H with a short circuit

Step 9 of 14



Step 10 of 14

In this circuit there is a short circuit across 20Ω resistor, therefore circuit reduced to



Step 11 of 14

$$R_{Th1} = 5\Omega$$

$$\tau_1 = \frac{2.5}{5}$$

$$\tau_1 = \frac{1}{2}$$

$$\therefore i_1(t) = 2.4e^{-\frac{t}{2}} \text{ A for } t > 0$$

$$\boxed{i_1(t) = 2.4e^{-2t} \text{ A}} \text{ for } t > 0$$

Step 12 of 14

Similarly second inductor current

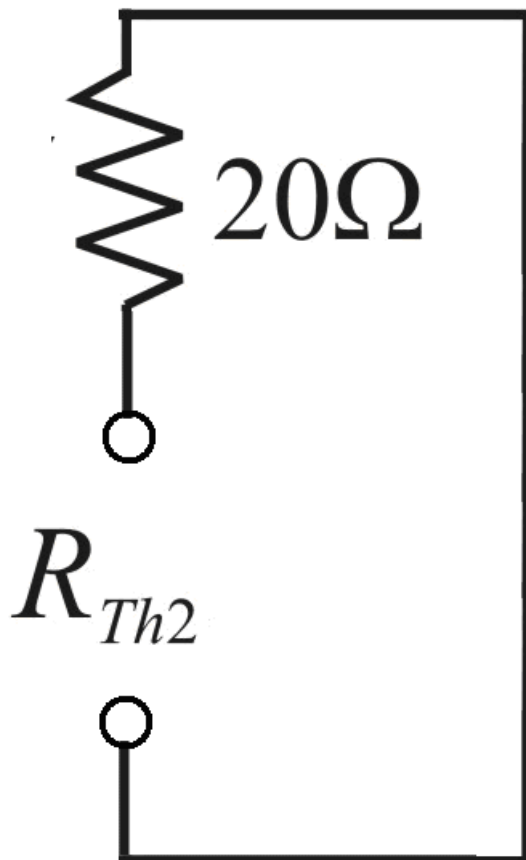
$$i_2(t) = i_2(0)e^{-\frac{t}{\tau_2}} \text{ for } t > 0$$

$$i_2(t) = 0.6e^{-\frac{t}{\tau_2}} \text{ for } t > 0$$

$$\text{Here } \tau_2 = \frac{L_2}{R_{Th2}}$$

Step 13 of 14

R_{Th2} is Thevenin resistance across inductor 4H by replacing inductor 2.5H with a short circuit



Step 14 of 14

$$R_{Th2} = 20\Omega$$

$$\tau_2 = \frac{4}{20}$$

$$\tau_2 = \frac{1}{5}$$

$$i_2(t) = 0.6e^{-\frac{t}{\frac{1}{5}}} \text{ for } t > 0$$

$$\boxed{i_2(t) = 0.6e^{-5t} \text{ A for } t > 0}$$